Lecture Notes

Lecture 13

Metric, Normed, and Hilbert Spaces

The core results of Lectures 1–12 used a standing finite-dimensional scope; their displayed infinite-dimensional spaces were previews. Quantum mechanics also uses infinite-dimensional function spaces, and to speak of convergence, limits, and complete orthonormal expansions there we need *analysis* layered on top of linear algebra. This lecture supplies it. We build metric and normed spaces, isolate the crucial property of *completeness* (Banach spaces), and arrive at *Hilbert spaces*—complete inner product spaces—with the two that run through all of physics, ℓ^2 and L^2. Orthonormal bases, Bessel's inequality, Parseval's identity, Fourier series, and the Riesz theorem all survive the passage to infinite dimensions, provided we respect completeness and continuity. This opens Part III.

Remark 13.1 (Hilbert-space scope and notation). This lecture allows real or complex Hilbert spaces. A Hamel basis gives finite algebraic expansions; a Hilbert basis is a complete orthonormal family and is countable only in a separable Hilbert space. From here $\mathcal{H}^*$ denotes the continuous dual. With the physics convention, Riesz is a conjugate-linear map $\mathcal{H} \to \mathcal{H}^*$ (equivalently, a linear map from the conjugate Hilbert space). General composite Hilbert spaces use completed, not merely algebraic, tensor products.

A note on rigor in Part III

Almost every result so far has been proved in full; the handful of exceptions (triangularizability and the generalized-eigenspace decomposition in Lecture 5, Jordan form in Lecture 6, and the perturbation results in optional Lecture 15) already carry the tags described here. Part III layers analysis onto linear algebra, and more of its results need more of it than this course assumes, so from here each theorem heading carries its status. An untagged result is proved here, as before; one marked sketched gives the central idea but omits a technical step (usually a convergence or completeness argument); and one marked cited is stated and motivated, with its full proof left to a functional-analysis course (Hall, in the syllabus references). For a cited result the goal is to know what it says and how to use it—not to reproduce the proof.

Learning objectives

By the end of this lecture you will be able to:

- Distinguish metrics, norms, and inner-product norms (via the parallelogram law).
- Define Cauchy sequences and completeness; tell Banach from Hilbert spaces.
- Work in the model spaces ℓ^2 and L^2 and compute Fourier expansions.
- Use a Hilbert basis with Bessel, Parseval, and completeness; state Riesz–Fréchet.

Remark 13.2 (Core, bridge, and extension route). The assessable core is the metric–norm–parallelogram chain; Cauchy sequences, completeness, and one completion; measure-aware ℓ^2 and L^2; Hilbert bases, Bessel, Parseval, projection, and bounded complex Riesz. One Fourier coefficient/approximation route, wavefunctions and number states, bounded multipliers, and the first unbounded domain form the bridge. The wider Fourier catalogue, absolute-series completeness criterion, orthogonal-polynomial catalogue, and rigged-space vista are extension material.

1 Metric and normed spaces

Definition 13.3 (Metric space). A *metric* on a set X is a function $d\colon X \times X \longrightarrow [0, \infty)$ with $d(x, y) = 0 \iff x = y$, $d(x, y) = d(y, x)$, and the triangle inequality $d(x, z) \le d(x, y) + d(y, z)$. The pair (X, d) is a *metric space*.

Definition 13.4 (Normed space). A *norm* on a vector space V assigns $\|v\| \ge 0$ with $\|v\| = 0 \iff v = 0$, $\|av\| = |a|\,\|v\|$, and $\|u + v\| \le \|u\| + \|v\|$. Every norm induces a metric $d(u, v) = \|u - v\|$.

Definition 13.5 (Convergence and Cauchy sequences). A sequence v_n *converges* to v if $\|v_n - v\| \longrightarrow 0$, and is *Cauchy* if for every $\varepsilon > 0$ there is N such that $\|v_m - v_n\| < \varepsilon$ whenever $m, n \ge N$. Convergent sequences are always Cauchy; the converse is the content of completeness.

Example 13.6 (Norms). On $\mathbb{C}^n$, for $n \ge 2$, the Euclidean norm $\|x\|_2 = \sqrt{\sum |x_i|^2}$ comes from the inner product, whereas the taxicab norm $\|x\|_1 = \sum |x_i|$ and the sup norm $\|x\|_\infty = \max_i |x_i|$ do not. In dimension one all three equal $|x|$ and are inner-product induced. On $C[a, b]$ the uniform norm $\|f\|_\infty = \sup|f|$ and the L^2 norm $\|f\|_2 = \sqrt{\int |f|^2}$ are genuinely different. ◁

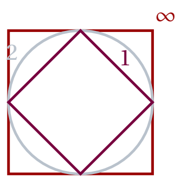

Figure 1: Unit balls in $\mathbb{R}^2$ for $\|\cdot\|_1$ (diamond), $\|\cdot\|_2$ (circle), and $\|\cdot\|_\infty$ (square). Only the round Euclidean ball comes from an inner product.

Example 13.7 (Cauchy, but where is the limit?). In $\mathbb{Q}$ the sequence $3,\ 3.1,\ 3.14,\ 3.141, \ldots$ is Cauchy, but its limit π is irrational: $\mathbb{Q}$ is incomplete. Filling the gaps gives $\mathbb{R}$. The same gap-filling turns an inner product space into a Hilbert space. ◁

Example 13.8 (Other metrics). Not every metric comes from a norm. The *discrete metric* $d(x, y) = 1$ for $x \ne y$ records only "same or different." For $1 \le p < \infty$, the standard p-norm is $\|x\|_p = (\sum |x_i|^p)^{1/p}$; $p = \infty$ means the sup norm. For dimension $n \ge 2$, only $p = 2$ is inner-product induced; for $n = 1$ all these norms equal $|x|$. When $0 < p < 1$ the same formula is a *quasi-norm*, not a norm: its triangle inequality can fail. ◁

Example 13.9 (Checking the triangle inequality). The triangle inequality is Cauchy–Schwarz in disguise: squaring, $\|u + v\|^2 = \|u\|^2 + 2\Re\langle u, v\rangle + \|v\|^2 \le \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$. The same one line proves it in every inner product space, finite- or infinite-dimensional. ◁

2 Completeness: Banach and Hilbert spaces

Definition 13.10 (Complete; Banach; Hilbert). A metric space is *complete* if every Cauchy sequence converges (to a point of the space). A complete normed space is a *Banach space*. An inner product gives a norm $\|v\| = \sqrt{\langle v, v\rangle}$; a complete inner product space is a *Hilbert space* $\mathcal{H}$.

Completeness is the assurance that limits one expects to exist actually do—there are no "missing points." Every finite-dimensional normed space is complete, so in finite dimensions the distinction never arose. It becomes essential in infinite dimensions, where natural spaces can have holes.

Example 13.11 (An incomplete inner product space). The continuous functions $C[-1,1]$ with $\langle f, g\rangle = \int_{-1}^{1} \overline{f} g$ form an inner product space that is *not* complete: continuous functions approximating a step are Cauchy in the L^2 norm yet converge to a discontinuous limit outside $C[-1,1]$. Its completion is $L^2[-1,1]$. ◁

Remark 13.12 (Which norms come from inner products). A norm comes from an inner product if and only if it obeys the parallelogram law $\|u+v\|^2+\|u-v\|^2 = 2\|u\|^2+2\|v\|^2$ (Lecture 7), in which case polarization recovers the inner product. (The forward direction is the computation of Lecture 7; the converse—that the parallelogram law forces the polarization formula to be an inner product—is the Jordan–von Neumann theorem, stated here without proof.) The sup norm fails it, so $C[a,b]$ with $\|\cdot\|_\infty$ is a Banach space but not a Hilbert space.

Proposition 13.13 (Optional extension: completeness via series · cited). A normed space is complete if and only if every absolutely convergent series (one with $\sum_n \|v_n\| < \infty$) converges. This is the working test for completeness, and the reason infinite expansions behave well in a Hilbert space.

Example 13.14 (Banach but not Hilbert). The sequence space ℓ^1 ($\sum|x_n| < \infty$, norm $\sum|x_n|$) is complete, hence Banach, but its norm violates the parallelogram law, so no inner product induces it. Among the ℓ^p spaces only ℓ^2 is a Hilbert space. ◁

Remark 13.15 (Finite dimensions are always complete). Every finite-dimensional normed space is complete, and all norms on it are equivalent, so the analytic subtleties of this lecture are invisible there. They surface only in infinite dimensions—exactly where quantum mechanics lives.

Remark 13.16 (The closed unit ball is no longer compact). A second, sharper symptom of infinite dimension: in $\mathbb{R}^n$ every closed bounded set is compact (every sequence in it has a convergent subsequence), but in an infinite-dimensional Hilbert space the closed unit ball fails this. The proof is one line: an orthonormal sequence $e_1, e_2, \dots$ lies in the unit ball, yet by the Pythagorean theorem

$$\|e_m - e_n\|^2 = \|e_m\|^2 + \|e_n\|^2 = 2 \qquad (m \neq n),$$

so all its terms stay a distance $\sqrt{2}$ apart and *no* subsequence can converge. "Closed and bounded" therefore no longer means compact, and the operators that retain finite-dimensional behavior will be exactly those that tame this failure—the *compact* operators of Lecture 14.

Example 13.17 (Completing a space). The polynomials on $[0,1]$ under the L^2 norm are incomplete, but they are dense (Weierstrass), so their completion is all of $L^2[0,1]$. Completion adds the limit of every Cauchy sequence—here, every square-integrable function. ◁

Example 13.18 (The sup norm fails the parallelogram law). Take $f = 1$ and $g(x) = x$ on $[0,1]$ in the sup norm: $\|f\|_\infty = \|g\|_\infty = 1$, $\|f+g\|_\infty = 2$, $\|f-g\|_\infty = 1$, so

$$\|f+g\|_\infty^2 + \|f-g\|_\infty^2 = 5 \neq 4 = 2\|f\|_\infty^2 + 2\|g\|_\infty^2.$$

The parallelogram law fails, confirming that the sup norm comes from no inner product. ◁

3 The Hilbert spaces of physics: ℓ^2 and L^2

Definition 13.19 (ℓ^2 and L^2). The sequence space ℓ^2 consists of $(x_1, x_2, \dots)$ with $\sum_n |x_n|^2 < \infty$, inner product $\langle x, y\rangle = \sum_n \overline{x_n} y_n$. The function space $L^2[a, b]$ consists of measurable functions f with finite Lebesgue integral $\int_a^b |f|^2 < \infty$, modulo equality almost everywhere; its inner product is $\langle f, g\rangle = \int_a^b \overline{f}\, g$. We quote this small amount of measure theory: the integral ignores changes on null sets, and the resulting quotient is complete.

Both are Hilbert spaces—the completeness of L^2 is the Riesz–Fischer theorem—and the standard ℓ^2 and interval $L^2[a, b]$ are *separable* (they admit a countable Hilbert basis). They are the state spaces of quantum mechanics: ℓ^2 for discrete systems (energy levels, spin chains), L^2 for a particle on a line (the position wavefunction $\psi(x)$).

Example 13.20 (Inside and outside ℓ^2). The sequence $(1, \frac{1}{2}, \frac{1}{3}, \dots)$ lies in ℓ^2 since $\sum 1/n^2 < \infty$, while the constant sequence $(1, 1, 1, \dots)$ does not. The standard vectors e_n (a 1 in slot n, zeros elsewhere) form a complete orthonormal basis, with $x = \sum_n x_n e_n$. ◁

Example 13.21 (Physics bridge: wavefunctions). A quantum state on the line is a $\psi \in L^2(\mathbb{R})$ with $\int |\psi|^2 = 1$, and $|\psi(x)|^2$ is a representative of its position probability density: probabilities of measurable regions are well-defined, although changing one point does not change the L^2 state. The Gaussian $\psi(x) = \pi^{-1/4} e^{-x^2/2}$, the oscillator ground state, lies in $L^2(\mathbb{R})$; plane waves e^{ikx} do not—a hint of the continuous spectrum to come. ◁

Remark 13.22 (Separability). ℓ^2 and L^2 are *separable*: each has a countable complete orthonormal basis, so every state is a countable superposition. All separable infinite-dimensional Hilbert spaces are *unitarily isomorphic after choosing a Hilbert basis*; they are not literally the same set or representation. Thus a chosen countable basis identifies either space with ℓ^2 through its coefficient map; for $L^2[-\pi, \pi]$ this is the Fourier-coefficient map.

Example 13.23 (Normalizing a state). The function $\psi(x) = e^{-|x|}$ on $\mathbb{R}$ has $\int_{-\infty}^{\infty} e^{-2|x|}\, dx = 1$, so it is already normalized in $L^2(\mathbb{R})$. Membership in L^2 together with norm one makes the Born position probability measure have total mass one. ◁

Example 13.24 (An L^2 inner product). On $L^2[0, 1]$, $\langle 1, x\rangle = \int_0^1 \overline{1} \cdot x\, dx = \frac{1}{2}$, while $\|1\|^2 = 1$ and $\|x\|^2 = \int_0^1 x^2 = \frac{1}{3}$. The functions 1 and x are not orthogonal—Gram–Schmidt will fix that. ◁

Example 13.25 (Cauchy–Schwarz for integrals). In L^2, Cauchy–Schwarz gives $|\int \overline{f} g| \le \|f\|_2 \|g\|_2$. Applying the same inequality to $|f|$ and $|g|$ also gives $\int |fg| \le \|f\|_2 \|g\|_2$. The second statement implies that the first integral is absolutely convergent whenever $f, g \in L^2$. ◁

Example 13.26 (Computing in ℓ^2). For $x = (1, \frac{1}{2}, \frac{1}{4}, \dots)$ and $y = (1, \frac{1}{3}, \frac{1}{9}, \dots)$, both in ℓ^2,

$$\langle x, y\rangle = \sum_{n\ge 0} \left(\tfrac{1}{2}\right)^n \left(\tfrac{1}{3}\right)^n = \sum_{n \ge 0} \left(\tfrac{1}{6}\right)^n = \frac{1}{1 - \frac{1}{6}} = \frac{6}{5}.$$

Geometric tails make ℓ^2 computations concrete. ◁

Example 13.27 (Normalizing on an interval). To normalize $\psi(x) = x$ on $[0, 1]$, compute $\|x\|^2 = \int_0^1 x^2\, dx = \frac{1}{3}$, so the normalized state is $\sqrt{3}\, x$. A constant on $[0, 1]$ normalizes to 1, the uniform state. ◁

Example 13.28 (Normalizing the Gaussian). The Gaussian $\phi(x) = e^{-x^2/2}$ has

$$\|\phi\|^2 = \int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi},$$

so the normalized oscillator ground state is $\pi^{-1/4}e^{-x^2/2}$, matching Example 13.21. Square-integrability is what keeps the normalization finite. ◁

Example 13.29 (Physics bridge: number states). The harmonic oscillator's energy eigenstates $|n\rangle$ ($n \geq 0$) are a complete orthonormal basis, so its Hilbert space is a copy of ℓ^2. A state $\sum_n c_n|n\rangle$ corresponds to the sequence (c_n), which is square-summable; the ladder action $a|n\rangle = \sqrt{n}\,|n-1\rangle$ defines an operator with natural dense domain

$$D(a) = \{(c_n)_{n\geq 0} \in \ell^2 : \sum_{n\geq 1} n|c_n|^2 < \infty\}.$$

It is densely defined and unbounded, the first domain-aware model for Lecture 14. ◁

Remark 13.30 (Any index set). One can form $\ell^2(\Omega)$ over any set Ω: families (x_ω) with $\sum|x_\omega|^2 < \infty$ (so at most countably many are nonzero). For countable Ω this is the separable ℓ^2; an uncountable Ω gives a non-separable Hilbert space, rare in physics but logically available.

4 Orthonormal bases in infinite dimensions

An indexed orthonormal family $(e_i)_{i\in I}$ is a *complete orthonormal basis* (a *Hilbert basis*) if its closed span is all of $\mathcal{H}$. Such a basis is countable exactly when $\mathcal{H}$ is separable. Completeness of the family (dense closed span) must not be confused with completeness of the ambient space (every Cauchy sequence has a limit there).

Remark 13.31 ("Basis" needs rethinking). The basis of Lecture 2—every vector a *finite* linear combination—is called a *Hamel basis*, and in an infinite-dimensional Hilbert space it becomes useless: one can show a Hamel basis of ℓ^2 or L^2 is necessarily uncountable, and no explicit one can be written down. The right notion replaces finite sums by norm-convergent sums: a Hilbert basis is an arbitrary indexed orthonormal family whose finite combinations come arbitrarily close to every vector (closed span = $\mathcal{H}$). The standard vectors e_n are a countable Hilbert basis of ℓ^2 but not a Hamel basis—the sequence $(1, \frac{1}{2}, \frac{1}{4}, \dots)$ is a limit of finite combinations, not a finite combination itself. From here on, "basis" in an infinite-dimensional Hilbert space always means Hilbert basis.

Theorem 13.32 (Expansion, Bessel, Parseval · sketched). Let $(e_i)_{i\in I}$ be an orthonormal family in a Hilbert space $\mathcal{H}$. For every v, Bessel's inequality holds,

$$\sum_{i\in I}|\langle e_i, v\rangle|^2 \leq \|v\|^2,$$

where the indexed sum is the supremum of its finite subsums. Bessel's inequality also shows precisely why each fixed vector has at most countably many nonzero coefficients: for every $m \geq 1$, only finitely many can have magnitude at least $1/m$, and the support is the countable union of those finite sets. The unordered net of finite Fourier sums converges in norm. The family is complete if and only if equality holds for all v (Parseval), equivalently $v = \sum_{i\in I}\langle e_i, v\rangle\, e_i$ for all v. Each vector has at most countably many nonzero coefficients, so its support may be enumerated to give a per-vector series even when I is uncountable. A single sequence indexing the whole basis is available exactly in the separable case.

Proof spine. For finite $F \subset I$, Pythagoras gives

$$\left\| v - \sum_{i \in F} \langle e_i, v \rangle e_i \right\|^2 = \|v\|^2 - \sum_{i \in F} |\langle e_i, v \rangle|^2 \geq 0,$$

which is Bessel. Hence coefficient tails make the finite Fourier sums Cauchy; Hilbert completeness supplies their norm limit s. Taking inner products with each e_i shows that $r = v - s$ is orthogonal to the closed span. Thus Parseval equality is equivalent to $\|r\| = 0$, and this is equivalent to completeness of the family and to the expansion $v = s$. ■

Theorem 13.33 (Characterizing a complete basis · cited). For an indexed orthonormal family $(e_i)_{i \in I}$ in a Hilbert space the following are equivalent: it is a complete orthonormal basis; it is maximal (no unit vector is orthogonal to every e_i); its closed span is the whole space; Parseval's identity holds for every vector.

The coefficients $\langle e_n, v \rangle$ are the Fourier coefficients, and the expansion is the abstract Fourier series of Lecture 8 carried to infinitely many terms; convergence is in the L^2 norm, not necessarily pointwise. The classical example is the trigonometric basis of $L^2[-\pi, \pi]$, whose expansion is the ordinary Fourier series.

Example 13.34 (Bridge: a Fourier series). The functions $\frac{1}{\sqrt{2\pi}} e^{inx}$ ($n \in \mathbb{Z}$) form a complete orthonormal basis of $L^2[-\pi, \pi]$. Expanding $f(x) = x$ gives $x = 2 \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \sin nx$ (convergence in L^2), and Parseval's identity applied to it reproduces $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$. ◃

Theorem 13.35 (Projection onto a closed subspace · cited). If M is a closed subspace of a Hilbert space $\mathcal{H}$, every v has a unique nearest point $Pv \in M$, with $v - Pv \perp M$; thus $\mathcal{H} = M \oplus M^\perp$ and P is the orthogonal projection.

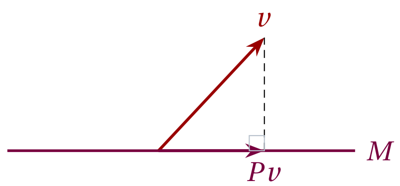

Figure 2: Projection onto a closed subspace: the nearest point Pv is the foot of the perpendicular from v, and $v - Pv$ is orthogonal to all of M. Closedness (hence completeness) is what guarantees the nearest point exists.

Example 13.36 (Truncation is the best approximation). Among all combinations of $e_1, \ldots, e_N$, the truncated Fourier sum $\sum_{n=1}^{N} \langle e_n, v \rangle\, e_n$ is the closest to v (Lecture 8, now in infinite dimensions). The squared error $\|v\|^2 - \sum_{n=1}^{N} |\langle e_n, v \rangle|^2$ shrinks to zero exactly when the basis is complete. ◃

Example 13.37 (A closed subspace and its complement). In $L^2[-1, 1]$ the even functions form a closed subspace M; its orthogonal complement $M^\perp$ is the odd functions, and every f splits uniquely as even-plus-odd, $f = \frac{1}{2}(f(x)+f(-x)) + \frac{1}{2}(f(x)-f(-x))$. This is $\mathcal{H} = M \oplus M^\perp$ in action. ◃

Example 13.38 (Optional extension: orthogonal-polynomial catalogue). Gram–Schmidt applied to monomials in $L^2(\mathbb{R}, e^{-x^2}dx)$ yields Hermite *polynomials*. Multiplying their normalized forms by $e^{-x^2/2}$ yields Hermite *functions*, a Hilbert basis of ordinary unweighted $L^2(\mathbb{R}, dx)$. The harmonic-oscillator energy basis is the Hermite-function basis, not the weighted polynomial family. (On $[-1,1]$ the analogous construction gives Legendre polynomials.) ◁

Remark 13.39 (Physics bridge: the Fourier transform). We use the unitary position–momentum convention

$$\widetilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} e^{-ipx/\hbar}\psi(x)\,dx, \qquad \psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} e^{ipx/\hbar}\widetilde{\psi}(p)\,dp.$$

Plancherel extends this transform to a unitary map on L^2. It carries the position representation to the momentum representation; writing wave number $k = p/\hbar$ instead moves the factor of $\hbar$ into that relation.

Example 13.40 (Optional extension: first Legendre polynomials). On $L^2[-1,1]$, Gram–Schmidt on $1, x, x^2$ gives (up to normalization) $p_0 = 1$ and $p_1 = x$ (already orthogonal to 1 since $\int_{-1}^{1} x = 0$). For the third, $\int_{-1}^{1} x^2 = \frac{2}{3}$ and $\int_{-1}^{1} 1 = 2$ give the component along 1 as $\frac{1}{3}$, so $p_2 = x^2 - \frac{1}{3}$. These are the Legendre polynomials. ◁

Remark 13.41 (Modes of convergence). Convergence in L^2 is weaker than pointwise: a Fourier series converges to f in the L^2 norm for every $f \in L^2$, yet may fail at individual points. For physics the L^2 sense is the relevant one—it is the convergence of probability amplitudes.

Example 13.42 (Optional extension: Parseval as an energy identity). For the Fourier basis, Parseval reads $\int |f|^2 = \sum_n |c_n|^2$ with c_n the Fourier coefficients: a signal's total energy is the sum of the energies of its frequency components. Applied to $f(x) = x$ on $[-\pi, \pi]$ it again gives $\sum 1/n^2 = \pi^2/6$. ◁

Example 13.43 (Bridge: a two-term Fourier approximation). The best approximation of $f(x) = x$ on $[-\pi, \pi]$ by $\{\sin x, \sin 2x\}$ is $2\sin x - \sin 2x$, its first two Fourier terms. Adding further terms only decreases the L^2 error, which tends to 0 because the trigonometric system is complete. ◁

Example 13.44 (Bridge: the selected Fourier coefficient). For $f(x) = x$ on $[-\pi, \pi]$ the coefficient against $\sin nx$ is

$$\frac{1}{\pi}\int_{-\pi}^{\pi} x \sin nx\,dx = \frac{2(-1)^{n+1}}{n},$$

evaluated by parts; summing recovers $x = 2\sum_{n\geq 1} \frac{(-1)^{n+1}}{n} \sin nx$. ◁

Example 13.45 (Bessel in action). For any v and orthonormal $e_1, \ldots, e_N$ the partial sum $\sum_{n=1}^{N} |\langle e_n, v\rangle|^2$ never exceeds $\|v\|^2$. Equality in the limit $N \to \infty$ is exactly completeness of the basis—no "energy" left unaccounted for. ◁

Example 13.46 (Projecting onto constants). The orthogonal projection of $f \in L^2[0,1]$ onto the constants is its average,

$$Pf = \frac{\langle 1, f\rangle}{\|1\|^2} = \int_0^1 f,$$

the best constant approximation. For $f(x) = x$ this is $\frac{1}{2}$, with residual $x - \frac{1}{2}$ orthogonal to the constants. ◁

Example 13.47 (Optional extension: a second Parseval sum). The square wave $\operatorname{sgn}(x)$ on $[-\pi,\pi]$ expands in odd sine harmonics, $\operatorname{sgn}(x) = \frac{4}{\pi}\sum_{k\ge 0}\frac{\sin(2k+1)x}{2k+1}$, and Parseval then gives $\sum_{k\ge 0}\frac{1}{(2k+1)^2} = \frac{\pi^2}{8}$. Different functions unlock different classical sums. ◁

Example 13.48 (Optional extension: energy of a sawtooth). For $f(x) = x$ on $[-\pi,\pi]$, direct integration gives $\|f\|^2 = \int_{-\pi}^{\pi} x^2\,dx = \frac{2\pi^3}{3}$. Parseval with the coefficients $\frac{2(-1)^{n+1}}{n}$ and $\|\sin nx\|^2 = \pi$ gives $\|f\|^2 = \pi\sum_{n\ge 1}\frac{4}{n^2}$; equating the two recovers $\sum 1/n^2 = \pi^2/6$ once more. ◁

Example 13.49 (Optional extension: a polynomial least-squares fit). The best linear approximation to $f(x) = x^2$ on $[-1,1]$—its projection onto $\operatorname{span}\{1,x\}$—uses $\langle 1,x^2\rangle = \frac{2}{3}$, $\langle 1,1\rangle = 2$, $\langle x,x^2\rangle = 0$:

$$Pf = \frac{\langle 1,x^2\rangle}{\langle 1,1\rangle}\,1 = \frac{1}{3}.$$

The residual $x^2 - \frac{1}{3}$ is the Legendre polynomial p_2 of Example 13.40, orthogonal to every linear function. ◁

Example 13.50 (Optional extension: Fourier series of $|x|$). The even function $|x|$ on $[-\pi,\pi]$ expands in cosines,

$$|x| = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k\ge 0}\frac{\cos(2k+1)x}{(2k+1)^2},$$

and Parseval reproduces $\sum_{k\ge 0}(2k+1)^{-4} = \pi^4/96$. Even functions use the cosine basis, odd functions the sine basis. ◁

Example 13.51 (Optional extension: building another Fourier series). For $f(x) = x^2$ on $[-\pi,\pi]$ the cosine coefficients are

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} x^2\,dx = \frac{\pi^2}{3}, \qquad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2\cos nx\,dx = \frac{4(-1)^n}{n^2},$$

so $x^2 = \frac{\pi^2}{3} + 4\sum_{n\ge 1}\frac{(-1)^n}{n^2}\cos nx$. Point evaluation is justified here: the coefficients are $O(n^{-2})$, so the Weierstrass M-test gives absolute uniform convergence to a continuous function. That limit agrees with x^2 almost everywhere by the L^2 expansion, hence everywhere because both are continuous. We may therefore set $x = \pi$ to recover $\sum 1/n^2 = \pi^2/6$, and $x = 0$ to get $\sum(-1)^{n+1}/n^2 = \pi^2/12$. ◁

Remark 13.52 (Optional extension: higher-dimensional bases). On the sphere the *spherical harmonics* $Y_{\ell m}$ are a complete orthonormal basis of $L^2(S^2)$; on $\mathbb{R}^3$ one multiplies them by radial functions. Expanding a wavefunction this way is separation of variables, and the labels ℓ, m are the angular-momentum quantum numbers—a complete set of commuting observables (Lecture 10).

5 The Riesz theorem in Hilbert space

A linear functional $\varphi\colon \mathcal{H} \to \mathbb{C}$ is *bounded* (equivalently continuous) if $|\varphi(v)| \le C\|v\|$ for some constant C.

Theorem 13.53 (Riesz–Fréchet · cited). Every bounded linear functional on a Hilbert space is $\varphi = \langle u,\cdot\rangle$ for a unique $u \in \mathcal{H}$, with $\|\varphi\| = \|u\|$. Under the physics convention, $u \mapsto \langle u,\cdot\rangle$ is a conjugate-linear isometric bijection $\mathcal{H} \to \mathcal{H}^*$, equivalently a linear isometry $\overline{\mathcal{H}} \to \mathcal{H}^*$ from the conjugate Hilbert space.

So the continuous dual is a conjugate-linear isometric copy of the space (the bra $\langle u|$ is the functional $\langle u, \cdot\rangle$), exactly as in Lecture 8—but only for the *bounded* functionals. In infinite dimensions unbounded functionals exist, and the restriction to continuous ones is the price of the infinite-dimensional dual. This continuity caveat returns with full force for operators in Lecture 14.

Example 13.54 (Bounded and unbounded functionals). On $L^2[0,1]$ the functional $f \mapsto \int_0^1 f$ is bounded (by Cauchy–Schwarz $|\int f| \le \|f\|_2$), so Riesz gives it as $\langle u, \cdot\rangle$ with $u \equiv 1$. Point evaluation $f \mapsto f(0)$ is not even well-defined on the L^2 quotient: representatives of one class may differ at 0. On the dense space $C[0,1]$ equipped with the L^2 norm it is well-defined but unbounded: the continuous spikes $f_n(x) = \max(1 - nx, 0)$ satisfy $f_n(0) = 1$ while $\|f_n\|_2^2 = 1/(3n) \to 0$. ◁

Example 13.55 (Bridge: a bounded multiplication operator). On $L^2[0,1]$, multiplication by $m \in L^\infty$, $(M_m f)(x) = m(x)f(x)$, is a bounded operator with $\|M_m\| = \operatorname{ess\,sup}|m|$, and its adjoint is multiplication by $\overline{m}$. A value changed at one point does not change the operator; an ordinary maximum agrees with the essential supremum for a continuous representative on a compact interval. Position $\hat{x}$ is the unbounded cousin $m(x) = x$ on all of $\mathbb{R}$, which is *not* bounded—the theme of Lecture 14. ◁

Example 13.56 (Adjoint of the shift). The right shift $S(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$ on ℓ^2 is bounded with $\|S\| = 1$, and its adjoint is the left shift

$$S^\dagger(x_1, x_2, \dots) = (x_2, x_3, \dots),$$

since $\langle Sx, y\rangle = \langle x, S^\dagger y\rangle$. Unlike in finite dimensions, S is an isometry that is not surjective—a purely infinite-dimensional phenomenon. ◁

Example 13.57 (Riesz in ℓ^2). The functional $\varphi(x) = \sum_n \frac{x_n}{n}$ on ℓ^2 is bounded, and Riesz identifies it with $u = (1, \frac{1}{2}, \frac{1}{3}, \dots) \in \ell^2$ (which lies in ℓ^2 since $\sum 1/n^2 < \infty$), because $\varphi(x) = \langle u, x\rangle = \sum_n \overline{u_n} x_n$. The functional and its representing vector carry the same data. ◁

Remark 13.58 (Optional extension: bras, kets, and rigged spaces). Kets are vectors in $\mathcal{H}$; bras are bounded functionals, with the conjugate-linear Riesz identification above. In the position representation, $\delta(x - x_0)$ represents an ideal position state and e^{ikx} an ideal momentum state. Neither belongs to $L^2(\mathbb{R})$; both are generalized vectors in an appropriate rigged-space dual, used to describe the continuous spectrum of Lecture 14.

Remark 13.59 (Bridge: toward unbounded operators). A linear map $T : \mathcal{H} \to \mathcal{H}$ is *bounded* if $\|Tv\| \le C\|v\|$, and then has an adjoint $T^\dagger$ defined just as in Lecture 9. But the operators of physics—position, momentum, energy—are typically *unbounded*, defined only on a dense domain. Handling them with care is the work of the final lecture.

Summary of main concepts

A norm supplies the metric $d(u, v) = \|u - v\|$; completeness turns normed spaces into Banach spaces and inner-product spaces into Hilbert spaces. The models are ℓ^2 and the almost-everywhere quotient L^2. Bessel holds for any orthonormal family, while a complete indexed family gives expansion and Parseval (a sequence expansion in the separable case). Riesz–Fréchet identifies a complex Hilbert space conjugate-linearly and isometrically with its *bounded* dual.

- **Metric and norm**—distance; a norm is a metric compatible with the linear structure, $d(u, v) = \|u - v\|$.
- **Completeness**—every Cauchy sequence converges; Banach (normed) and Hilbert (inner product); automatic in finite dimensions.
- **ℓ^2 and L^2**—square-summable sequences and square-integrable functions—the state spaces of quantum mechanics.
- **Hilbert basis**—an arbitrary indexed complete orthonormal family; countable exactly in the separable case; Parseval for a basis and Bessel for any orthonormal family.
- **Riesz–Fréchet**—$u \mapsto \langle u, \cdot \rangle$ is a conjugate-linear isometric bijection onto the bounded dual.
- **Why infinite dimensions differ**—the closed unit ball is no longer compact.

Exercises for this lecture are in the companion sheet `exercises-13.pdf`.